LESSON 12 · CHEAT SHEET

SQL II: Across the Shelves

JOIN (INNER)

```
SELECT h.name, q.monster, p.loot_share
FROM party AS p
JOIN heroes AS h ON p.hero_id = h.hero_id
JOIN quests AS q ON p.quest_id = q.quest_id
WHERE q.outcome = 'Success'
-- only rows that match on both sides
```

LEFT JOIN

```
SELECT q.region,
       COALESCE(r.warden, '(none)')
FROM quests AS q
LEFT JOIN regions AS r
       ON q.region = r.region
WHERE r.region IS NULL  -- the unmatched
-- filter on r? put it in ON, not WHERE
```

CASE WHEN

```
CASE WHEN reward >= 1000 THEN 'Rich'  -- big
     WHEN reward >= 400  THEN 'Fair'  -- first!
     ELSE 'Small'
END AS band
SUM(CASE WHEN outcome = 'Success'
         THEN 1 ELSE 0 END)  -- COUNTIF
```

SUBQUERIES

```
WHERE level > (SELECT AVG(level) FROM heroes)
WHERE hero_id NOT IN (SELECT hero_id FROM ...)
-- per row: the inner query sees the outer
WHERE q.reward > (SELECT AVG(q2.reward)
                  FROM quests AS q2
                  WHERE q2.monster = q.monster)
```

WITH (A NAMED STEP)

```
WITH loot AS (
    SELECT hero_id, SUM(loot_share) AS total
    FROM party GROUP BY hero_id
)
SELECT h.name, l.total
FROM loot AS l
JOIN heroes AS h ON l.hero_id = h.hero_id
ORDER BY l.total DESC LIMIT 5
```

VALUES FROM PYTHON: ?

```
sql = "SELECT * FROM quests WHERE region = ?"
con.execute(sql, (region,))   # tuple: comma!
con.execute(sql2, (region, 3))  # two ?s
# NEVER: f"... WHERE region = '{region}'"
```

INTO PANDAS

```
import pandas as pd
df = pd.read_sql_query(sql, con)
df = pd.read_sql_query(sql, con, params=(region,))
```

EXCEL → PANDAS → SQL

EXCEL	PANDAS	SQL
XLOOKUP	`merge(how="inner")`	`JOIN`
XLOOKUP + #N/A	`merge(how="left")`	`LEFT JOIN`
ISNA	`.isna()`	`IS NULL`
IFERROR	`.fillna(x)`	`COALESCE`
IF / IFS	`pd.cut`	`CASE WHEN`
COUNTIF	`(s == x).sum()`	`SUM(CASE ...)`
helper cell	a variable	subquery / `WITH`

GOTCHAS

- JOIN with no ON: every row times every row
- Same column in two tables? Prefix it: `p.quest_id`
- INNER JOIN drops the unmatched without a word
- `= NULL` is never true: `IS NULL`
- CASE: first true WHEN wins
- Row-per-hero joins repeat quest numbers
- `(region)` is a string; `(region,)` is a tuple